

# Browser Credential Gathering - Linux Privilege Escalation

## Introduction

Browser credential gathering is a powerful yet often overlooked technique for privilege escalation. If you gain access to a user's home directory, it is possible to extract saved credentials, cookies, and session tokens from popular browsers like Chromium and Firefox. These credentials can provide direct access to sensitive systems, online accounts, or private data, making this method both effective and alarming.

Misconfigured systems or insecure credential storage practices amplify this risk. This cheatsheet outlines how to extract credentials, tools that streamline the process, and best practices for securing browsers.

## 1. Credential Gathering Basics

Saved browser credentials are typically stored in encrypted databases or files. If decryption is possible, attackers can extract sensitive information.

| Steps                              | Description                                                                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Locate browser storage directories | Check common paths: - Chromium/Chrome: ~/.config/google-chrome/ or ~/.config/chromium/ - Firefox: ~/.mozilla/firefox/                |
| Identify profiles                  | For Chromium/Chrome, look in the Default or Profile X directories. For Firefox, locate the active profile directory in profiles.ini. |

| Steps                  | Description                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| Extract database files | Look for files such as: - Login Data (Chromium/Chrome) - logins.json (Firefox) - cookies.sqlite (session tokens). |

## 2. Extracting Credentials

### Chromium/Chrome Credential Extraction

| Steps                  | Description                                                                              |
|------------------------|------------------------------------------------------------------------------------------|
| Locate Login Data file | Found in ~/.config/google-chrome/[Profile]/Login Data.                                   |
| Use decryption tools   | Tools like <a href="#">SharpChromium</a> can decrypt and extract credentials.            |
| Manual method          | Use sqlite3 to read the database and decrypt the passwords using the Linux user keyring. |

### Firefox Credential Extraction

| Steps                          | Description                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------|
| Locate logins.json and key4.db | Found in ~/.mozilla/firefox/[Profile]/.                                                    |
| Decrypt credentials            | Use <a href="#">firefox_decrypt</a> to extract saved credentials.                          |
| Extract session tokens         | Query cookies.sqlite using sqlite3 to retrieve authentication cookies for active sessions. |

### Example: Extracting Chrome Saved Passwords

| Step                       | Command/Description                                            |
|----------------------------|----------------------------------------------------------------|
| Export Login Data file     | cp ~/.config/google-chrome/Default/Login\ Data /tmp/login_data |
| Open database with sqlite3 | sqlite3 /tmp/login_data                                        |

| Step                  | Command/Description                                                     |
|-----------------------|-------------------------------------------------------------------------|
| List tables           | sqlite> .tables                                                         |
| Extract login details | sqlite> SELECT origin_url, username_value, password_value FROM logins;  |
| Decrypt passwords     | Use a tool or script to decrypt passwords using the Linux user keyring. |

### 3. Tools for Browser Credential Gathering

| Tool            | Description                                                                       | Link                                 |
|-----------------|-----------------------------------------------------------------------------------|--------------------------------------|
| SharpChromium   | Extracts Chromium/Chrome credentials, cookies, and autofill data.                 | <a href="#">GitHub</a>               |
| firefox_decrypt | Decrypts Firefox saved credentials using the corresponding key files.             | <a href="#">GitHub</a>               |
| sqlite3         | General-purpose tool for querying browser databases (Login Data, cookies.sqlite). | Pre-installed on most Linux systems. |

### 4. Securing Browsers Against Credential Theft

To protect systems and users against browser credential theft, consider the following measures:

| Security Practice             | Description                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------|
| Disable credential storage    | Use password managers instead of saving credentials in browsers.                  |
| Encrypt sensitive directories | Encrypt user directories with tools like eCryptfs to prevent unauthorized access. |
| Restrict file permissions     | Ensure that browser storage directories are accessible only by the user.          |

| Security Practice   | Description                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------|
| Monitor file access | Use tools like auditd to monitor access to sensitive files such as Login Data and cookies.sqlite. |

## 5. Additional Techniques

### Session Hijacking via Cookies

Cookies often contain session tokens, which can be used to hijack active sessions.

| Steps                                                 | Description                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------|
| Locate cookies.sqlite (Firefox) or Cookies (Chromium) | Found in the browser profile directories.                                  |
| Extract session cookies                               | Query the database for cookies related to login or authentication.         |
| Use cookies in requests                               | Import cookies into your browser or tools like curl for session hijacking. |

| Command                                                                          | Description                  |
|----------------------------------------------------------------------------------|------------------------------|
| sqlite3 ~/.mozilla/firefox/[Profile]/cookies.sqlite "SELECT * FROM moz_cookies;" | List all cookies in Firefox. |

## Key Notes

- Browser credential gathering is a powerful escalation vector because it can bypass multi-factor authentication (MFA) using session tokens.
- Always ensure proper hygiene of browser credentials by minimizing saved passwords and monitoring access to sensitive files.
- These techniques highlight the importance of robust system and browser security practices.